

Computational study of the covalent bonding of microcystins to cysteine residues – a reaction involved in the inhibition of the PPP family of protein phosphatases

Susana R. Pereira¹, Vítor M. Vasconcelos^{1,2} and Agostinho Antunes^{1,2}

¹ CIIMAR/CIMAR, Centro Interdisciplinar de Investigação Marinha e Ambiental, Universidade do Porto, Portugal

² Departamento de Biologia, Faculdade de Ciências, Universidade do Porto, Portugal

Keywords

computational study; covalent bond; microcystins; PPP family; protein phosphatases

Correspondence

S. R. Pereira and A. Antunes, CIIMAR, Centro Interdisciplinar de Investigação Marinha e Ambiental, Rua dos Bragas 289, 4050-123 Porto, Portugal
Fax: +351 223 390 608
Tel: +351 223 401 837; +351 223 401 813
E-mails: spereira@ciimar.up.pt; aantunes@ciimar.up.pt
Website: <http://www.ciimar.org/CIIMAR/>

(Received 7 September 2011, revised 10 November 2011, accepted 6 December 2011)

doi:10.1111/j.1742-4658.2011.08454.x

Microcystins (MCs) are cyclic peptides, produced by cyanobacteria, that are hepatotoxic to mammals. The toxicity mechanism involves the potent inhibition of protein phosphatases, as the toxins bind the catalytic subunits of five enzymes of the phosphoprotein phosphatase (PPP) family of serine/threonine-specific phosphatases: Ppp1 (aka PP1), Ppp2 (aka PP2A), Ppp4, Ppp5 and Ppp6. The interaction with the proteins includes the formation of a covalent bond with a cysteine residue. Although this reaction seems to be accessory for the inhibition of PPP enzymes, it has been suggested to play an important part in the biological role of MCs and furthermore is involved in their nonenzymatic conjugation to glutathione. In this study, the molecular interaction of microcystins with their targeted PPP catalytic subunits is reviewed, including the relevance of the covalent bond for overall inhibition. The chemical reaction that leads to the formation of the covalent bond was evaluated *in silico*, both thermodynamically and kinetically, using quantum mechanical-based methods. As a result, it was confirmed to be a Michael-type addition, with simultaneous abstraction of the thiol hydrogen by a water molecule, transfer of hydrogen from the water to the α,β -unsaturated carbonyl group of the microcystin and addition of the sulfur to the β -carbon of the microcystin moiety. The calculated kinetics are in agreement with previous experimental results that had indicated the reaction to occur in a second step after a fast noncovalent interaction that inhibited the enzymes *per se*.

Introduction

Microcystins (MCs) are cyclic heptapeptides that are produced by several genera of cyanobacteria and are hepatotoxic to mammals. MCs cannot move across cell membranes [1], but are able to leave the digestive tract and enter hepatocytes through active uptake by a non-specific organic anion-transport system [2]. In the mammalian liver, MCs can be conjugated with reduced glutathione (GSH), a general detoxification mechanism

that enhances the water solubility of xenobiotic compounds in order to aid their renal excretion. GSH-conjugated MCs have been identified in several aquatic organisms (fish, mussels, daphnids and macrophytes) and in rodents [3,4]. They are somewhat less toxic than nonconjugated MCs [5,6].

The cellular mechanism of MC toxicity involves the potent inhibition of protein phosphatases present in

Abbreviations

Adda, 2S,3S,8S,9S,4E,6E-3-amino-9-methoxy-2,6,8-trimethyl-10-phenyl-4,6-decadienoic acid; GSH, reduced glutathione; Masp, D-methyl-aspartate; MC, microcystin; Mdha, N-methyl-dehydro-alanine; PPP, phosphoprotein phosphatase.

the cytoplasm [7]. More explicitly, the toxins are known to interact with the catalytic subunits of five enzymes of the phosphoprotein phosphatase (PPP) family of serine/threonine-specific phosphatases: Ppp1 (aka PP1), Ppp2 (aka PP2A), Ppp4, Ppp5 and Ppp6. The metazoan PPP family is composed of seven enzymes, but the activity of the remaining two – Ppp3 (aka PP2B) and Ppp7 (aka PPEF) – is not affected by MCs [8].

Over one-third of all proteins undergo reversible phosphorylation at one or more residues, a chemical modification that modulates their conformation, activity, localization and/or stability [9,10]. As protein phosphatases catalyze the removal of the phosphate group from those residues, they are key players for maintaining an adequate level of protein phosphorylation at every moment of the cellular cycle and it would not be an overstatement to say that they control virtually every physiological process and every cell-signaling pathway. The PPP family of serine/threonine-specific phosphatases has the additional particularity of being the physiological target of a number of natural toxins, of which MCs are a classic example [8]. The understanding of the chemical interaction between PPP enzymes and the toxic compounds will help to predict possible harmful effects and to explore their pharmacological potential in greater detail.

In this study, the molecular interaction of MCs with their targeted PPP catalytic subunits was characterized in detail, and particular emphasis was given to a covalent bond that is known to be formed between a PPP cysteine residue and the methylene group of an MC. The reaction mechanism of the covalent addition was clarified through *in silico* calculations using quantum mechanical-based methods. The small size of the models used in these calculations allows the generalization of the results to other biological systems that involve this specific type of reaction. This may prove interesting because the addition of cysteine residues to the methylene group of MCs seems to have a more general occurrence, having been suggested to be involved in the biological role of the toxins [11]. Moreover, the same chemical reaction is responsible for the nonenzymatic conjugation of MCs with GSH.

Microcystins

MCs are synthesized by nonribosomal enzyme complexes [12,13] and contain unusual, nonproteinogenic amino acids (see Fig. 1). The seven constituent amino acids are (a) D-alanine (D-Ala), (b) a variable X residue, (c) D-methyl-aspartate (Masp), (d) a variable Z residue, (e) (2*S*,3*S*,8*S*,9*S*,4*E*,6*E*)-3-amino-9-methoxy-2,6,8-

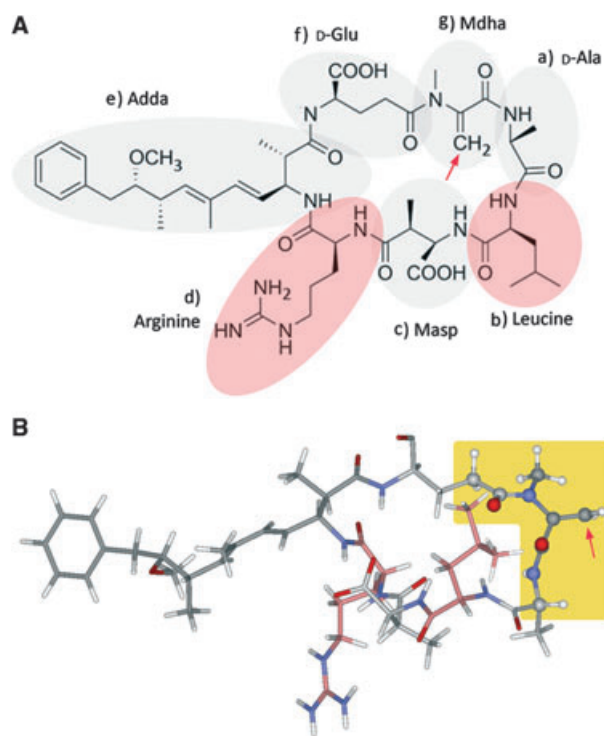


Fig. 1. Chemical structure of MCs. MCs are cyclic heptapeptides with five common and two variable amino acid residues. (A) Chemical structure of MCLR, where the variable residues, highlighted in pink, correspond to a leucine and an arginine. (B) 3D structure of MCLR; the leucine and arginine carbon atoms are represented in pink; and the atoms shown as a ball-and-stick model with the yellow background correspond to those included in the computational study. The red arrow indicates the methylene group where the cysteine residue binds covalently.

trimethyl-10-phenyl-4,6-decadienoic acid (Adda), (f) D-glutamic acid (D-Glu) and (g) *N*-methyl-dehydro-alanine (Mdha).

Over 80 MCs have been identified so far, presenting diverse X and/or Z residues, and/or minor differences in the other amino acids. The variable X and Z residues are always L-amino acids that are indicated in the name of the molecule by their one-letter code (e.g. MCLR has leucine and arginine in positions 2 and 4, respectively). The modifications in the other amino-acid residues include demethylation of Masp and Mdha, and methylesterification of D-Glu. The different MCs have different toxicity profiles, the most toxic being MCLR [14]. MCLR has similar potency in the inhibition of Ppp1, Ppp2, Ppp4, Ppp5 and Ppp6, presenting a half maximal inhibitory concentration (IC_{50}) of around 0.1–1 nM [15–18]. It is 1000-fold less potent towards Ppp3 (IC_{50} of $\sim 1 \mu M$) and seems to have no effect on the activity of Ppp7 [18,19].

Interaction of MCs with PPP catalytic subunits

There are three available crystallographic structures of PPP catalytic subunits with bound MCs: Ppp1c α with MCLR [20], Ppp1c γ with 2H-MCLA [21] and Ppp2c α with MCLR [22]. 2H-MCLA differs from MCLR in that it presents an alanine in the place of arginine as the variable residue **Z**, and an *N*-methyl-alanine instead of Mdha, given the saturation of the double bond provided by the two extra hydrogen atoms.

From these structures, it is evident that MCs bind directly in the catalytic center of the PPP enzymes, completely blocking the access of the substrate to the active site. The area of the catalytic subunits where the toxins bind presents three surface grooves – the hydrophobic groove, the acidic groove and the C-terminal groove – which occur in a Y-shape with the active site situated at the bifurcation point (see Fig. 2).

The Adda side-chain of MCs is accommodated closely in the hydrophobic groove and accounts for a significant portion of the toxin's binding potential – modifications in the configuration of the Adda side-chain that produce a bent orientation unable to accommodate as tightly reduce the potency of the toxin [23,24]. The D-Glu (carboxyl group) of MCs, as well as the adjacent carbonyl group, make hydrogen bonds to metal-bound water molecules, which are important interactions as D-Glu was found to be essential for the toxicity of MC [24,25]. Additionally, the carboxyl group of Masp is hydrogen-bonded to two PPP conserved residues: an arginine and a tyrosine (Arg96 and Tyr134 of Ppp1c; and Arg89 and Tyr127 of Ppp2c).

A final interaction site consists of the PPP β 12– β 13 loop, where the S γ atom of a cysteine residue (Cys273 in Ppp1c and Cys269 in Ppp2c) bonds covalently to the Mdha of MCLR, and the aromatic ring of a conserved tyrosine (Tyr272 of Ppp1c; and Tyr265 of Ppp2c) packs closely to the leucine residue of MCs, suggesting the presence of hydrophobic interactions.

Comparing the three structures, there is one clear structural change in the toxin-binding area, which only occurs in the case of Ppp1c bound to MCLR. It consists of a shift in the β 12– β 13 loop that results from the formation of the covalent bond with Cys273 (see Fig. 2). This shift is not observed in the structure of 2H-MCLA-bound Ppp1c because the hydrogenation of the Mdha residue precludes the formation of the covalent bond. In Ppp2c bound to MCLR the loop is not shifted because the cysteine residue involved in the bond is Cys269 (Phe276 in Ppp1c) and not Cys266 (corresponding to Ppp1c Cys273).

The covalent bond of MCs to cysteine residues

Interestingly, whereas all the PPP catalytic subunits that are inhibited by MCs have at least one cysteine residue in the β 12– β 13 loop in position to form the covalent bond with the Mdha methylene group, those that are resistant to MC (Ppp3 and Ppp7) do not have a cysteine residue available at this position (see Fig. 2). Ppp4c and Ppp6c have two cysteine residues corresponding to Cys266 and Cys269 in Ppp2c; Ppp5 has only one cysteine residue, corresponding to the Cys273 residue of Ppp1c (Ppp2c Cys266), indicating that its interaction with MCs will also result in the shift of the β 12– β 13 loop.

Although the previous observation is unlikely to be just a coincidence, the formation of the covalent bond is not considered essential for the inactivation of the enzymes by MCs [26]. Several findings support this conclusion: the existence of equally potent PPP-inhibiting MCs that cannot form the covalent bond (e.g. Asp, ADMAdda, Dhb-microcystin-HtyR) [27]; the existence of other naturally occurring, PPP-inhibiting toxins that do not bind them covalently (e.g. nodularins, which are cyanobacteria-produced pentapeptidic toxins structurally related to MCs); the finding that MCLR inhibits Ppp1 and Ppp2 in a two-step mechanism, and, while the noncovalent interaction completely inactivates the enzymes within minutes, the formation of the covalent bond between the cysteine and the Mdha residues takes hours [28]; and the fact that glutathione-conjugated MCs (or cysteine-conjugated MCs) still have the capacity to inhibit PPP activity, although with less potency [5].

Nevertheless, even if the formation of the covalent bond is accepted as an accessory event in the context of PPP inhibition, it should no doubt contribute to the stabilization of the enzyme–toxin complex. Additionally, the reaction appears to have relevance outside this context, as it has been proposed to be involved in the, so far elusive, biological role of MCs [11] and is responsible for the nonenzymatic conjugation of MCs to GSH. In fact, whereas conjugation with GSH is usually catalyzed by a glutathione *S*-transferase it can also occur spontaneously given strong alkaline conditions [3,4].

The nature of the chemical reaction itself is generally believed to consist of a Michael-type addition of the nucleophilic sulfur atom of the cysteine residue to the α,β -unsaturated carbonyl group of the Mdha residue, but so far this has not been confirmed. In this study, quantum mechanical-based methods were used to clarify the chemical mechanism of the formation of the

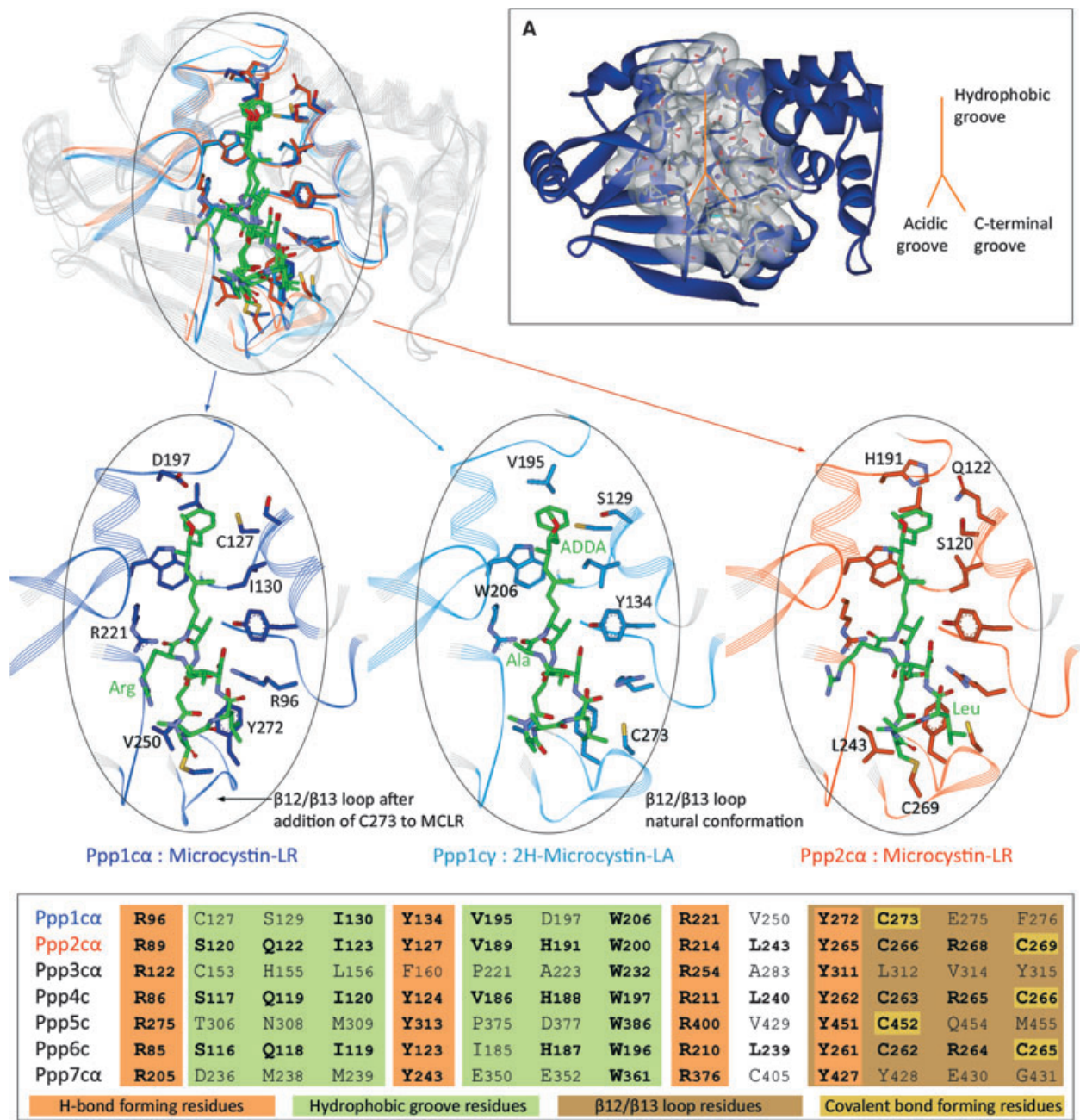


Fig. 2. Global and detailed view of the interaction of MCs with PPP catalytic subunits. Three crystallographic structures of microcystin-bound PPP catalytic subunits are represented superimposed and in individual close-ups of the toxin-binding site (PDB ID: [1fjm](#), [2bdx](#) and [2ie3](#)). (A) The PPP surface area where the toxins bind have three grooves arranged in a Y configuration, with the active site situated at its bifurcation. The partial sequence alignment shown at the bottom of the figure includes the Ppp1c and Ppp2c amino acid residues that interact with MCs – as observed in the referred crystallographic structures – and the corresponding residues in the other PPP enzymes. Ppp3c and Ppp7c are the only PPP catalytic subunits with no covalent bond-forming cysteine residues in the β12–β13 loop; coincidentally, they also offer higher resistance to inhibition by MCs.

covalent bond and to ascertain the energetics involved in the reaction.

The three optimized geometries of the stationary points corresponding to the reaction that was found to

be more favorable, both thermodynamically and kinetically, are represented in Fig. 3. This reaction represents the addition of the cysteine sulfur atom to the β-carbon of the α,β-unsaturated carbonyl moiety of

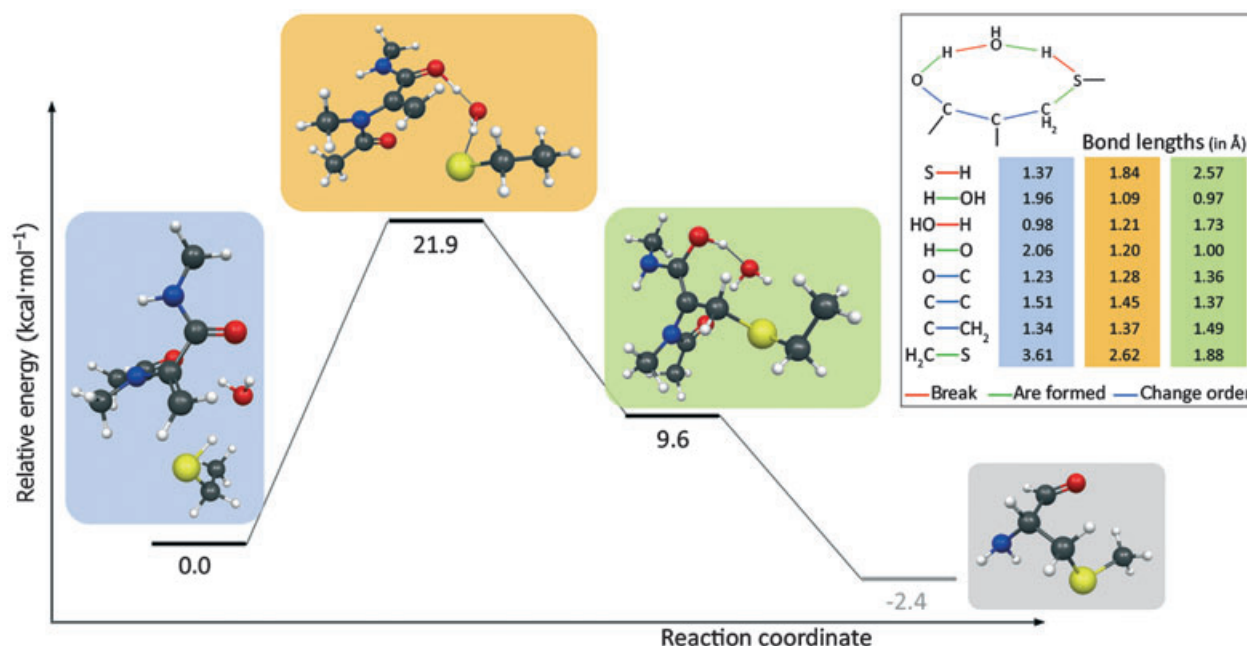


Fig. 3. Stationary points of the modeled reaction, representing the addition of a cysteine residue to the methylene group of MCs. The modeled system includes part of an MC, the side-chain of the cysteine residue and a water molecule. The reaction coordinate was found to be complex: with the addition of the sulfur atom to the carbon of the MC, the hydrogen of the thiol group passes to the water, which releases one of its hydrogen atoms to form a covalent bond with the oxygen of the Mdha carbonyl group. As the inserted panel shows, there are a total of eight covalent bonds undergoing prominent changes: two that break, three that are formed, and three that change order. The relative energy of the ketonic form of the product has been extrapolated from calculations with smaller models.

the Mdha residue, with the assistance of a water molecule. The presence of the water as a catalyst was found to be necessary, as calculations with alternative reaction coordinates, where no water was present, involved prohibitive energies and/or resulted in thermodynamically unstable products (data not shown).

The reaction coordinate is complex: with the addition of the sulfur atom to the carbon, the hydrogen of the thiol group passes to the water, which releases one of its hydrogen atoms to form a covalent bond with the oxygen of the Mdha carbonyl group. There are a total of eight covalent bonds involved in the reaction, which undergo prominent changes: two that break; three that are formed; and three that change order (see the inset in Fig. 3). In the end-product, the sulfur is covalently bonded to the carbon of an enolic compound, which should decay to the corresponding ketone as this form seems to be thermodynamically more favorable.

The classic Michael reaction refers to the addition of a carbanion to the β -carbon of an α,β -unsaturated carbonyl compound, but the concept has been generalized to include nucleophiles such as sulfur, nitrogen, phosphorous or oxygen. The assistance of a base is usually required to activate the nucleophilic attack, which is

consistent both with the alkaline medium needed to trigger the nonenzymatic conjugation of GSH with MCs and with the role played by the water molecule in these calculations.

The calculated activation energy for this reaction is $21.9 \text{ kcal}\cdot\text{mol}^{-1}$ (see Fig. 3), which, using transition state theory, corresponds to a rate constant of around 10^{-3} s^{-1} ($2.4\text{E } 10^{-3} \text{ s}^{-1}$ for $T = 310.15 \text{ K}$ and $1.0\text{E } 10^{-3} \text{ s}^{-1}$ for $T = 303.15 \text{ K}$). If the reaction is regarded as pseudo first-order, by considering the MC concentration constant in relation to the protein, the corresponding half-life for $T = 303.15 \text{ K}$ is approximately 2 h. This value is in agreement with the previous finding that the formation of the covalent bond between the cysteine and the Mdha residues was the second step of the interaction of MCLR with Ppp1 and Ppp2, which took place a couple of hours after a first noncovalent interaction completely inactivated the enzymes [28].

These calculations do not contribute to the understanding of the differences in the toxicity of diverse MC isoforms, or their toxicity towards diverse PPP enzymes. It would be interesting to compare MCs with different variable residues but we believe that the influence of those residues is more likely to reside in the

noncovalent part of the toxin–enzyme interaction [8] and would not have an impact on the results obtained in this study.

In conclusion, the computational study of the reaction mechanism that is responsible for the formation of a covalent bond between the Mdha residue of MCs and a cysteine residue, confirmed the reaction to be a Michael-type addition, with simultaneous abstraction of the thiol hydrogen by a water molecule, addition of the sulfur to the Mdha methylene carbon and transfer of the hydrogen from the water to the Mdha carbonyl oxygen. The calculated kinetics are in agreement with previous experimental results.

The importance of characterizing this reaction is primarily related to the toxicological properties of MCs, which involve interaction with protein phosphatases of the PPP family. However, the insights obtained can also be applied both to the ecological role of these toxins, proposed to include binding of a variety of proteins in response to oxidative stress, and to the nonenzymatic conjugation of the toxin to GSH, which is thought to constitute the first step in the biotransformation of the molecule.

Methods

Density functional theory (DFT), at the B3LYP/6-31G(d) level of theory [29,30], was used within the General Atomic and Molecular Electronic Structure System (GAMESS) software package [31] to perform all the calculations. The model system had a total of 35 atoms: those from MCLR are represented as a ball-and-stick model in Fig. 1B, plus the side-chain of the cysteine residue added, plus a water molecule.

The computational study of each reaction consisted of the following steps. First, the transition state was found by optimizing a close geometry, obtained from a scan of the reaction coordinate, and was confirmed through the identification of the respective imaginary frequency. Second, intrinsic reaction coordinate calculations were performed to identify the two minima connected by that transition state. MacMolPlt graphical user interface [32] was used to visualize the geometric and electronic features of the different stationary points.

Acknowledgements

S.R.P. was supported by grant SFRH/BPD/26639/2006 from the Portuguese Foundation for Science and Technology (FCT). This work was also partly funded by FCT projects PTDC/AAC-AMB/104983/2008 (FCOMP-01-0124-FEDER-008610) and PTDC/AAC-CLI/116122/2009 (FCOMP-01-0124-FEDER-014029).

References

- 1 Eriksson JE, Grönberg L, Nygård S, Slotte JP & Meriluoto JAO (1990) Hepatocellular uptake of 3H-dihydromicrocystin-LR, a cyclic peptide toxin. *Biochim Biophys Acta* **1025**, 60–66.
- 2 Runnegar M, Berndt N & Kaplowitz N (1995) Microcystin uptake and inhibition of protein phosphatases: effects of chemoprotectants and self-inhibition in relation to known hepatic transporters. *Toxicol Appl Pharmacol* **134**, 264–272.
- 3 Pflugmacher S, Wiegand C, Oberemm A, Beattie KA, Krause E, Codd GA & Steinberg CEW (1998) Identification of an enzymatically formed glutathione conjugate of the cyanobacterial hepatotoxin microcystin-LR: the first step of detoxication. *Biochim Biophys Acta* **1425**, 527–533.
- 4 Takenaka S (2001) Covalent glutathione conjugation to cyanobacterial hepatotoxin microcystin LR by F344 rat cytosolic and microsomal glutathione S-transferases. *Environ Toxicol Pharmacol* **9**, 135–139.
- 5 Kondo F, Ikai Y, Oka H, Okumura M, Ishikawa N, Harada KI, Matsuura K, Murata H & Suzuki M (1992) Formation, characterization, and toxicity of the glutathione and cysteine conjugates of toxic heptapeptide microcystins. *Chem Res Toxicol* **5**, 591–596.
- 6 Metcalf JS, Beattie KA, Pflugmacher S & Codd GA (2000) Immuno-crossreactivity and toxicity assessment of conjugation products of the cyanobacterial toxin, microcystin-LR. *FEMS Microbiol Lett* **189**, 155–158.
- 7 Campos A & Vasconcelos V (2010) Molecular mechanisms of microcystin toxicity in animal cells. *Int J Mol Sci* **11**, 268–287.
- 8 Pereira SR, Vasconcelos VM & Antunes A (2011) The phosphoprotein phosphatase family of Ser/Thr phosphatases as principal targets of naturally occurring toxins. *Crit Rev Toxicol* **41**, 83–110.
- 9 Cohen P (2000) The regulation of protein function by multisite phosphorylation – a 25 year update. *Trends Biochem Sci* **25**, 596–601.
- 10 Hunter T (1995) Protein-kinases and phosphatases – the yin and yang of protein-phosphorylation and signaling. *Cell* **80**, 225–236.
- 11 Zilliges Y, Kehr JC, Meissner S, Ishida K, Mikkat S, Hagemann M, Kaplan A, Börner T & Dittmann E (2011) The cyanobacterial hepatotoxin microcystin binds to proteins and increases the fitness of *Microcystis* under oxidative stress conditions. *PLoS ONE* **6**, e17615, doi:10.1371/journal.pone.0017615.
- 12 Tillett D, Dittmann E, Erhard M, Von Döhren H, Börner T & Neilan BA (2000) Structural organization of microcystin biosynthesis in *Microcystis aeruginosa* PCC7806: an integrated peptide-polyketide synthetase system. *Chem Biol* **7**, 753–764.

- 13 Nishizawa T, Ueda A, Asayama M, Fujii K, Harada KI, Ochi K & Shirai M (2000) Polyketide synthase gene coupled to the peptide synthetase module involved in the biosynthesis of the cyclic heptapeptide microcystin. *J Biochem* **127**, 779–789.
- 14 Zurawell RW, Chen HR, Burke JM & Prepas EE (2005) Hepatotoxic cyanobacteria: a review of the biological importance of microcystins in freshwater environments. *J Toxicol Environ Health B Crit Rev* **8**, 1–37, doi:10.1080/10937400590889412.
- 15 Chen MX, McPartlin AE, Brown L, Chen YH, Barker HM & Cohen PTW (1994) A novel human protein serine/threonine phosphatase, which possesses 4 tetratricopeptide repeat motifs and localizes to the nucleus. *EMBO J* **13**, 4278–4290.
- 16 Hastie CJ & Cohen PTW (1998) Purification of protein phosphatase 4 catalytic subunit: inhibition by the antitumour drug fostriecin and other tumour suppressors and promoters. *FEBS Lett* **431**, 357–361.
- 17 Prickett TD & Brautigan DL (2006) The alpha 4 regulatory subunit exerts opposing allosteric effects on protein phosphatases PP6 and PP2A. *J Biol Chem* **281**, 30503–30511.
- 18 Swingle M, Ni L & Honkanen RE (2007) Small molecule inhibitors of ser/thr protein phosphatases: specificity, use and common forms of abuse. *Methods Mol Biol* **365**, 23–38.
- 19 Huang XZ & Honkanen RE (1998) Molecular cloning, expression, and characterization of a novel human serine/threonine protein phosphatase, PP7, that is homologous to Drosophila retinal degeneration C gene product (rdgC). *J Biol Chem* **273**, 1462–1468.
- 20 Goldberg J, Huang HB, Kwon YG, Greengard P, Nairn AC & Kuriyan J (1995) Three-dimensional structure of the catalytic subunit of protein serine/threonine phosphatase-1. *Nature* **376**, 745–753.
- 21 Maynes JT, Luu HA, Cherney MM, Andersen RJ, Williams D, Holmes CFB & James MNG (2006) Crystal structures of protein phosphatase-1 bound to motuporin and dihydromicrocystin-LA: elucidation of the mechanism of enzyme inhibition by cyanobacterial toxins. *J Mol Biol* **356**, 111–120, doi:10.1016/j.jmb.2005.11.019.
- 22 Xing Y, Xu YH, Chen Y, Jeffrey PD, Chao Y, Lin Z, Li Z, Strack S, Stock JB & Shi YG (2006) Structure of protein phosphatase 2A core enzyme bound to tumor-inducing toxins. *Cell* **127**, 341–353, doi:10.1016/j.cell.2006.09.025.
- 23 Harada KI, Ogawa K, Matsuura K, Murata H, Suzuki M, Watanabe MF, Itezono Y & Nakayama N (1990) Structural determination of geometrical isomers of microcystins LR and RR from cyanobacteria by two-dimensional NMR spectroscopic techniques. *Chem Res Toxicol* **3**, 473–481.
- 24 An J & Carmichael WW (1994) Use of a colorimetric protein phosphatase inhibition assay and enzyme linked immunosorbent assay for the study of microcystins and nodularins. *Toxicon* **32**, 1495–1507.
- 25 Namikoshi M, Rinehart KL, Sakai R, Stotts RR, Dahlem AM, Beasley VR, Carmichael WW & Evans WR (1992) Identification of 12 hepatotoxins from a homer lake bloom of the cyanobacteria *Microcystis aeruginosa*, *Microcystis viridis*, and *Microcystis wesenbergii*: nine new microcystins. *J Org Chem* **57**, 866–872.
- 26 MacKintosh RW, Dalby KN, Campbell DG, Cohen PTW, Cohen P & MacKintosh C (1995) The cyanobacterial toxin microcystin binds covalently to cysteine-273 on protein phosphatase 1. *FEBS Lett* **371**, 236–240.
- 27 Hastie CJ, Borthwick EB, Morrison LF, Codd GA & Cohen PTW (2005) Inhibition of several protein phosphatases by a non-covalently interacting microcystin and a novel cyanobacterial peptide, nostocyclin. *Biochim Biophys Acta* **1726**, 187–193, doi: 10.1016/j.bbagen.2005.06.005.
- 28 Craig M, Luu HA, McCready TL, Williams D, Andersen RJ & Holmes CFB (1996) Molecular mechanisms underlying the interaction of motuporin and microcystins with type-1 and type-2A protein phosphatases. *Biochem Cell Biol* **74**, 569–578.
- 29 Stephens PJ, Devlin FJ, Chabalowski CF & Frisch MJ (1994) Ab Initio calculation of vibrational absorption and circular dichroism spectra using density functional force fields. *J Phys Chem* **98**, 11623–11627.
- 30 Becke AD (1993) Density-functional thermochemistry. III. The role of exact exchange. *J Chem Phys* **98**, 5648–5652.
- 31 Schmidt MW, Baldrige KK, Boatz JA, Elbert ST, Gordon MS, Jensen JH, Koseki S, Matsunaga N, Nguyen KA, Su S *et al.* (1993) General atomic and molecular electronic structure system. *J Comp Chem* **14**, 1347–1363.
- 32 Bode BM & Gordon MS (1998) Macmolplt: a graphical user interface for GAMESS. *J Mol Graph Model* **16**, 133–138.